FSKU: Open GPU Compute Indexing & Forward Curves

A Transparent, Public Methodology for Normalized Spot Benchmarks, Price Dispersion, and Model-Implied Forward Term Structures

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1. Executive Summary & Market Problem

As accelerated compute becomes the foundational capital expenditure of modern artificial intelligence, market participants require transparent, liquid, and mathematically reproducible benchmarks. Historically, GPU cloud pricing has remained opaque, dominated by unlisted negotiated rates, disparate multi-GPU chassis configurations, and fragmented pricing terms across hyperscalers (AWS, GCP, Azure) and specialized neoclouds (CoreWeave, RunPod, Lambda Labs).

Commercial data providers have historically responded to this opacity by selling paywalled, proprietary indices. However, closed-box pricing indices create severe institutional hazards: methodology opacity, unverifiable source feeds, and conflation between surface-level quotes and executed trades. **FSKU** solves this by establishing a 100% open-source, reproducible benchmark pipeline: *Public Data In → Normalized $/GPU-hr → SKU Spot Index → Historical Benchmark → Implied Forward Curve*.

2. Unit Normalization Architecture

Cloud compute is sold across heterogeneous server topographies: single-GPU instances, 8-GPU HGX clusters, and multi-node reservations. To establish true like-for-like comparability across providers without introducing arbitrary subjective modifiers, FSKU applies strict mathematical **Unit Normalization**:

```
Normalized Rate ($/GPU-hour) = Published Hourly Server Rate / Published Physical GPU Count
```

Normalization Boundaries: Normalized rates establish unit parity ($/GPU-hr) rather than full economic identity. No opaque synthetic multipliers are added for CPU allocations, system RAM, NVMe scratch disks, or egress terms. Every normalized rate is explicitly categorized by its primary **Contract Basis**: *On-demand*, *Spot*, *Capacity Block*, or *Retail API*.

3. Benchmark Index Construction Methodologies

To protect benchmarks against extreme upper-tail retail quotes or promotional spot distortions, FSKU evaluates six statistical aggregation formulas:

Methodology	Formula / Specification	Institutional Application & Purpose
Robust Median (Default)	`S_0 = Median(P_1, P_2, ..., P_n)`	Default anchor. Highly resilient against extreme upper-tail enterprise retail API quotes.
10% / 20% Trimmed Mean	`Mean(P[k : N-k]), k = floor(N * p)`	Symmetrically removes highest and lowest outliers to produce an insulated arithmetic mean.
Provider-Balanced	`Mean(Median_Provider(P))`	Equal-weights each provider first before averaging, eliminating provider quote-density bias.
GPU-Count Weighted	`Sum(Total_Rate) / Sum(GPU_Count)`	Weights index by physical cluster scale (e.g. 8-GPU HGX cluster price vs 1-GPU PCIe price).
Simple Arithmetic Mean	`1/N * Sum(P_i)`	Baseline reference. Highly vulnerable to upward skew in distributed retail cloud markets.

4. Matched-Provider Technological Deflation (d)

A critical flaw in naive compute forward pricing is assuming historical hardware retains static rental value. In reality, rapid silicon architecture iteration causes steady rental price deflation across older chip generations. To infer pure hardware deflation (**d**) without contaminating the estimate with provider markup variations, FSKU calculates rental ratios strictly across **matched provider pairs** offering consecutive GPU architectures (e.g. A100 → H100, H100 → H200, H100 → B200):

```
Annual Technological Deflation (d) = 1 - ( P_older / P_newer ) ^ ( 12 / Cadence_Months )
```

5. Model-Implied Forward Term Structure Formulation

Forward curves in FSKU are explicitly model-implied term structures calculated dynamically from current cash spot anchors, inferred technological deflation (**d**), and a configurable annual carry & scarcity factor (**c**):

```
F(T) = S_0 * [ (1 + c) * (1 - d) ] ^ T   |   where T = m / 12 (delivery horizon in years)
```

Term Structure Dynamics: When annual tech deflation (d ≈ 18%) exceeds annual scarcity/carry (c = 5%), the forward curve displays natural backwardation (decay roll), reflecting the empirical reality of chip aging. Interquartile ranges (Q25 & Q75) provide clear confidence bands across 1M, 3M, 6M, 12M, 24M, 36M, 48M, and 60M delivery tenors.

6. Methodology Sensitivity & Source Ablation Resilience

Institutional indices must withstand source volatility and provider dropouts. FSKU incorporates two real-time diagnostic engines:

- **Methodology Sensitivity:** Computes the maximum divergence spread across all six weighting formulas to quantify outlier vulnerability.
- **Source Ablation:** Systematically drops each provider from the dataset and calculates the resulting index delta (`Delta %`) to reveal single-source concentration risk.

Empirical Snapshot: Benchmark Indices Across Key Accelerator SKUs

GPU SKU	VRAM	Spot Index	10% Trimmed	Provider-Bal.	12M Forward	36M Forward	Status
H100 SXM	80 GB	**$2.49**	$3.26	$3.16	$2.15	$1.60	ACTIVE
H200	141 GB	**$6.30**	$7.23	$7.93	$5.44	$4.06	ACTIVE
B200	180 GB	**$5.98**	$6.28	$6.21	$5.16	$3.85	EMERGING
B300	288 GB	**$5.71**	$5.71	$5.71	$4.93	$3.68	EMERGING
A100 SXM	80 GB	**$1.29**	$1.32	$1.34	$1.11	$0.83	ACTIVE
MI300X	192 GB	**$2.40**	$2.40	$2.40	$2.07	$1.55	INDICATIVE

7. Software Architecture & Open Governance

FSKU is engineered as a zero-dependency, high-throughput compute intelligence platform in Python 3.10+ with:

- **Embedded NoSQL Storage (FSKUDb):** Thread-safe document collections with atomic persistence and SHA-256 snapshot audits.
- **Resync Engine:** Asynchronous multi-provider polling adapters with automated diff detection.
- **FastAPI REST Server & Rich CLI:** High-performance JSON endpoints, CSV streaming exports, and terminal visualization.
- **Apache 2.0 License:** Permissive, patent-protected open-source licensing to ensure universal enterprise and research adoption.

Repository & Reference Code: https://github.com/nativ-x/fsku | Built by: NATIVX (nativx.net) | Inquiries: research@nativx.net